RAMA COMPUTER & NETWORK SOLUTION

Web Portal, SLA Maintenance & System Operations Guide

1. EXECUTIVE OVERVIEW

RAMA Computer and Network Solution operates as an integrated IT infrastructure provider in Addis Ababa, Ethiopia, bridging active hardware procurement, certified structured cabling, and proactive Annual Maintenance Contracts (AMCs) under a single operational umbrella. This user guide outlines the architecture of the 5-page enterprise web portal, the underlying SQLite database schema, the REST API endpoints, and the operational protocols executed by the Network Operations Center (NOC) and field engineering teams.

2. WEB PORTAL ARCHITECTURE & NAVIGATION MAP

The RAMA web portal consists of five interconnected HTML pages styled with modern, responsive layouts tailored for local commercial clients in Bole, Kazanchis, Kirkos, Arada, and Lideta:

Page File	Target Purpose & Core Offerings	Key Form / Action
index.html	Main landing hub highlighting RAMA's 'one-stop-shop' value proposition and local reliability badges.	Free 30-min Site Health Audit Booking
hardware-maintenance.html	Phase 1 focus page detailing active gear procurement, passive cabling, and UPS power resilience.	Parts & AMC Quotation Request
sla-support-tiers.html	Structured SLA maintenance packages (Gold, Silver, Bronze) with a full comparison matrix.	SLA Onboarding Enrollment Form
equipment-catalog.html	Certified hardware catalog linking sub-pages for Switches, Cables, Racks, and UPS systems.	Bulk Parts Order Quote
about.html	Company story, CCNA/MikroTik engineering certifications, and target industry sector profiles.	Consultation Booking

3. SLA MAINTENANCE TIERS & RESPONSE PROTOCOLS

RAMA structures its Annual Maintenance Contracts (AMCs) into three defined tiers to ensure predictable operating expenditure (OpEx) for local organizations:

SLA Tier	P1 Response Window	NOC Monitoring	Preventive Audits	Hardware Swap Pool
Gold SLA	< 2 Hours Priority	24/7/365 Continuous	Bi-Weekly Visits	Reserved Hot-Swap Pool
Silver SLA	< 4 Hours Guaranteed	12/5 Business Hours	Monthly Audits	Standard Local Pool
Bronze SLA	Next Business Day	Helpdesk Ticketing	Quarterly Audits	Out-of-Warranty RMA

4. BACKEND DATABASE & REST API ARCHITECTURE

The backend system utilizes SQLite (rama_sla_database.db) managed through a Flask REST API (sla_rest_api.py). The system consists of four primary relational tables:

• **sla_tiers:** Catalog table storing tier definitions, P1/P2 response time thresholds, and base rates in USD and ETB.

• **client_sectors:** Reference table categorizing client accounts into SME, Banking, Healthcare, Hospitality, and Real Estate.

• **sla_inquiries:** Core transactional table storing lead submissions, company contacts, district locations, and estimated device counts.

• **audit_schedules:** Operational table managing assigned Cisco/MikroTik field engineers, audit dates, and Fluke/UPS test results.

> **REST API Endpoints Summary:**
> • **POST /api/sla/submit:** Ingests web form submissions, auto-generates tracking codes (e.g., RAMA-SLA-2026-0001), and creates an initial audit dispatch entry.
> • **GET /api/sla/inquiry/<ref>:** Returns live contract status, SLA tier parameters, and scheduled engineer visit times.
> • **POST /api/engineer/update-ticket:** Enables field technicians in Addis Ababa to log Fluke cable test and UPS voltage pass/fail results directly from client sites.

5. FIELD TECHNICIAN EMERGENCY DISPATCH PROTOCOL

When a Priority 1 emergency ticket is dispatched, field technicians must follow a 4-phase protocol:
1. **Pre-Dispatch Preparation:** Confirm SLA tier response window, load latest switch/firewall configuration backups, and pack the emergency field kit (Fluke DSX tester, SFP transceivers, temporary PoE swap switch).
2. **On-Site Triage & Isolation:** Measure rack PDU input voltage, verify UPS battery load, inspect active gear LED status, and capture console logs.
3. **Resolution & Swap Execution:** Apply soft patches or mount a temporary replacement switch from RAMA's local swap pool.
4. **Client Sign-off:** Perform post-repair Fluke channel verification, demonstrate connectivity, and log client verification in the mobile API.